Athmeeya M Kashyap

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Education

IIIT Hyderabad 2024 - 2029 (Expected)
B.Tech Computer Science + MS Computational Natural Sciences *Hyderabad, India*

- Coursework: Probability & Random Processes, Linear Algebra, Statistical Mechanics, Machine & Data Learning, Discrete Mathematics, Data Structures & Algorithms.

Projects

Heston Stochastic-Volatility Options Pricer with Adjoint Greeks | *C++, Python, PyTorch, NumPy*

- Priced options under the **Heston** model two ways, Fourier inversion and Monte Carlo, that agree to within Monte Carlo error and both collapse to **Black-Scholes** at constant volatility (to 10^{-5}).
- Fit the model to a live S&P 500 option chain and reproduced the **volatility smile** flat Black-Scholes cannot, cutting implied-vol error from **4.65** to **0.76** vol points (**6.2x**).
- Computed every Greek in one **reverse-mode (adjoint)** pass, **3.9x** faster than bumping each input separately and accurate to the closed-form Black-Scholes Greeks.
- Trained a **neural network** to return price and all 9 Greeks in a single **18.7-microsecond** pass, **278x** faster than the full pricer at under **1%** out-of-sample error, on the C++ engine I built for **ForgeLM**.

No-Regret Market-Making Engine | *C++17, Python, pybind11*

- Built an **adaptive market maker** that re-finds the best quote in **1 round** after a sudden regime change, versus **161** for a static learner: a **161x** faster recovery.
- Verified its **no-regret guarantee** empirically: cumulative regret held at **63%** of the theoretical worst-case bound over **10,000** rounds, so it provably never trails the best fixed quote by much.

GTO Poker Solver | *Python, CFR, Bayesian Inference*

- Developed and compared **4** counterfactual-regret (CFR) solvers on Kuhn and Leduc poker; the best drove **exploitability** to **0.0008** chips per hand, **33%** below vanilla CFR at the same **10,000** iterations.
- Added an **opponent model** that detects deviations from optimal play and switches from defensive to exploitative, tying poker bluff frequency to how bid-ask spreads form in markets (**Glosten-Milgrom**).

Research Experience

Undergraduate Researcher, CCNSB Lab, IIIT Hyderabad Jan 2026 - Present
Advisor: Prof. Chittaranjan Hens | Complex Networks, Dynamical Systems & Phase Transitions *Hyderabad, India*

Flash-Crash Dynamics in Coupled-Oscillator Networks | *Python, C++, SciPy*

- Modeled a market-wide crash as a **desynchronization transition** on a coupled-oscillator network; the simulated onset of collapse matched the analytical critical coupling to within **2%**.
- Showed that **variance and lag-1 autocorrelation** rise sharply just before the collapse, producing an **early-warning signal** of an imminent crash.

Random Graphs & Financial Contagion | *Python, NetworkX, SciPy*

- Wrote a percolation generator that recovers the Erdos-Renyi critical exponents **exactly** (-1.500 and $-1/3$) across **10,000+** graphs and scales to **1,000,000-node** networks.
- Implemented three interbank-contagion models (percolation, threshold cascade, DebtRank) on **1,000-node** networks, showing **scale-free hub networks** fail at far lower stress than evenly-connected ones.

Achievements & Leadership

INMO Qualifier (2022): Qualified for the Indian National Mathematical Olympiad (among the top ∼300 nationally).

INMOTC (2022): Selected for INMO Training Camp at ICTS-TIFR, Bengaluru.

NSEA, Top 1% Karnataka: Top 1% statewide in the National Standard Examination in Astronomy (IAPT).

Math Olympiad Instructor, CFAL Mangaluru: Teach the four olympiad domains (algebra, geometry, combinatorics, number theory) for national olympiad preparation.

SGFI National Swimming: Represented Karnataka in Swimming at the School Games Federation of India Nationals.

Technical Skills

Quantitative Methods: Probability, Statistical Inference, Stochastic Processes, Stochastic Volatility (Heston), Monte Carlo, Derivatives Pricing & Calibration, Adjoint Algorithmic Differentiation, Game Theory (CFR/Nash), Network Analysis

Machine Learning: Deep Learning, Neural Networks, LLM Inference & Serving, PyTorch, Scikit-learn

Languages: Python, C++, C, Golang, SQL, Bash

Scientific Computing: NumPy, SciPy, Pandas, NetworkX, Matplotlib, pybind11

Systems & Infrastructure: Git, Linux, Docker, PostgreSQL, AWS